

Character Swap Studio — Workflow Guide

A friend's guide to two flows: the **Swap workflow** (turn a scene + a person into finished, captioned videos) and the **Multi-clip editor workflow** (stitch a pile of takes into one clean reel). Written for someone who has the app in front of them.

0. One-time setup

```
cd ~/character-swap-workflow
~/local/bin/uv sync                # install Python deps
cp .env.example .env              # then edit .env (see keys below)
~/local/bin/uv run character-swap serve # opens http://127.0.0.1:8000
```

Required keys (in `.env`) — the core flows won't run without these:

Key	Unlocks
<code>OPENAI_API_KEY</code>	Image generation (GPT Image) and Whisper transcription (captions / pacing)
<code>XAI_API_KEY</code>	Grok video + Grok images

Optional keys — each unlocks more models/features (locked ones show  in the picker):

Key	Unlocks
<code>GEMINI_API_KEY</code>	Nano Banana / Nano Banana Pro (best swap image quality), Veo 3 video
<code>KLING_ACCESS_KEY</code> + <code>KLING_SECRET_KEY</code>	Kling video models incl. Kling 3.0 (need <i>both</i> keys)
<code>ANTHROPIC_API_KEY</code>	 AI Director (Claude writes tailored prompts from the actual images)
<code>ELEVENLABS_API_KEY</code>	Voice swap / TTS
<code>HEYGEN_API_KEY</code>	Talking-head avatars (Avatar tab)

Optional — animated captions: for the 4 modern Remotion templates (submagic-pro, mrbeast-bold, capcut-glow, capcut-purple-pill), run once (needs Node ≥ 18):

```
~/local/bin/uv run character-swap remotion-install
```

Without it, the 19 legacy (ASS) caption templates still work.

Tip — keep your data safe: by default `state/` `characters/` `input/` `output/` live inside the working folder. Point `STATE_DIR` / `CHARACTERS_DIR` / `INPUT_DIR` / `OUTPUT_DIR` at a shared `~/character-swap-data/` (and symlink `.env`) so nothing gets wiped. The server is localhost-only and `.env` is git-ignored — keep real keys there.

1. The Swap workflow

Goal: drop in scene background(s) + a character, generate a photorealistic swap, animate each shot, and compile one finished MP4 per character.

The Swap tab walks through numbered steps. You can leave and come back — jobs survive browser/server restarts, and milestone notifications (🔔 in the header) ping you when images are ready.

Step 1 — Scene(s)

Drop one or more background images (drag, click, or **Cmd+V** paste). Each image is a separate scene; every character gets variants for **every** scene. A counter shows (N scenes — each character gets variants for every scene).

Step 2 — Character images

- Pick one or more characters from the library (upload new ones inline).
- **Model picker:** default is GPT Image 2. **Nano Banana Pro is the recommended pick for swaps** — strongest at keeping the scene/pose and relighting the person so they don't look pasted in.
- **Images per character** (1–4): how many variants to generate per scene to choose from.
- Optional **prompt** edit, plus two quality toggles:
- ✨ **enrich** — cheap GPT-4o prompt expansion (only when you typed a short custom prompt).
- 🎬 **AI Director** — Claude reads the actual reference images and writes a tailored prompt per (character × scene). Slower/pricier (~\$0.05) but worth it for consistency. Greyed out unless `ANTHROPIC_API_KEY` is set.

The default prompt already enforces **Option B** (keep the scene's real background exactly), full identity override, prop/layout lock, exact framing, and **lighting integration** so the person blends into the scene's light. You usually don't need to touch it.

Step 3 — Generate & approve

Variants appear grouped by character (and scene). For each (character × scene), click ✓ on the one you like (click again to un-approve). Use ✓ **Approve all** to auto-pick the first ready one everywhere. Per-variant you also get: 🛠 **edit** (branch a variant with a tweak prompt), ↺ **retry** a failed slot, ↓ **download**.

If an image gets refused for "content policy": the app auto-retries once with a tiny softening clause, and as a last resort re-runs that variant on Nano Banana Pro (a different moderation backend) — so borderline-but-tasteful shots usually still go through. Nothing for you to do.

Step 3.5 — Arrange scenes (the ⇅ panel)

Once you've approved at least one variant, an **Arrange scenes** strip appears. This is where you build the final video sequence (the "slot" model):

- 📄 **Duplicate** a scene → another clip from the *same image*, with its own motion prompt + duration (great for multiple beats from one shot).
- ⬅ / ➡ reorder scenes (sets the order they're stitched in Step 6).
- ✕ **Delete** a scene slot (can't delete the last one).

Step 4 — Movement prompt (per scene)

One **motion prompt + duration per scene**, shared by all that scene's images:

- Write what should happen ("he pours the bottle and looks at camera, says '...'").
- **Duration** per scene (e.g. 5s / 10s), and **Video model** for the whole

job — Grok Imagine, Veo, **Kling (incl. 3.0)**, Runway, Luma, etc. - **Videos per approved image** (1–10). - Hit ► **Animate**. (Submitting here **locks** Steps 1–4 — approvals and scenes freeze. Use "Unlock movement" if no video has finished yet.)

Step 5 — Videos

The chosen model animates each approved image. Live progress + per-video download.

Step 6 — Compile

Once a character has ≥ 1 finished video, **Compile** concatenates that character's scenes (in your arranged order) into one MP4 and runs the Editor pipeline: silence trim → optional voice swap (per-character preset 🗣️ voice or a batch override) → Whisper → WPM normalize → caption burn-in. All characters compile in parallel.

Shortcut — already have finished images? Use the 🗑️ **Animate tab** instead of Steps 1–3: drop your finished images in order, and you jump straight to per-scene motion prompts → videos → compile.

2. The Multi-clip editor workflow

Goal: you have several takes (HeyGen/Higgsfield/recorded), possibly out of order. Paste the script they follow and the app transcribes, reorders, paces, stitches, and captions them into one clean reel.

Needs `OPENAI_API_KEY` (Whisper). Voice swap needs `ELEVENLABS_API_KEY`. Animated caption templates need the one-time `remotion-install`.

Step 1 — Open the Editor tab → Multi-clip mode

Editor tab → toggle "**Multi-clip (auto-order)**" at the top.

Step 2 — Paste the full script

Paste the complete text the clips were meant to follow. The app fuzzy-matches each clip's transcript to a position in the script (difflib), so **clip upload order doesn't matter** — the script is the source of truth for final order. ~15% similarity is enough; weakly-matched clips are flagged and appended at the end.

Step 3 — Add the clips

Drag-drop multiple videos, click **+ add clips**, or import recent renders from the **Higgsfield Inbox** (one click). Any format ffmpeg reads works. Video-only clips (no audio) are placed by script position.

Step 4 — Set the options

- **Trim silences** (on) — removes gaps. (*A per-clip leading-silence trim always runs so every clip starts exactly on the first spoken word.*)
- **Captions** (on) — word-level captions burned in.
- **Normalize pace to [WPM]** (on, default 190) — time-stretches **each clip independently** so all takes land at the same speaking pace. Pitch-preserving.
- **Speed** (0.75x – 2x, default 1x) — a global pitch-preserving speed-up of the whole finished reel, applied on top of WPM. Captions stay in sync.

- **Voice swap** (optional) — pick an ElevenLabs voice to replace all audio.
- Expand **Caption template + style** to choose a template (capcut-purple-pill is the default) and tweak size/colors/position.

Step 5 — Stitch + auto-edit

Click the big **Stitch + auto-edit** button. Pipeline: transcribe each clip → match + reorder → per-clip WPM normalize → leading-trim → concat → interior trim → optional voice swap → global speed-up → captions. ~1–3 min depending on clip count.

Step 6 — Review & fine-tune

The result panel shows: - **Match order** — where each clip landed + its match score (red = unmatched). - **Pace normalization** — per-clip WPM decision (↑ sped up, ↓ slowed down). -  **Edit captions** — a visual editor: scrubbing timeline + draggable word cards + per-word text/timing edits, with live preview. -  **Rerender** — swap caption template/style without re-transcribing (fast). - **Trim & split** — a CapCut-style timeline to trim/split/reorder segments.

Quick gotchas

- **Captions greyed out?** `OPENAI_API_KEY` is missing (Whisper).
- **A model shows**  ? Its API key isn't in `.env` (hover for which one).
- **Remotion caption templates missing?** Run `character-swap remotion-install` (Node ≥ 18), then restart.
- **Can't edit approvals/scenes anymore?** You already hit ► Animate in Step 4 — it locks. Start a new job or "Unlock movement" (only before a video finishes).
- **A swapped person looks pasted-in?** Use **Nano Banana Pro** + the default prompt (it has the lighting-integration directives), or turn on  AI Director.
- **Very high speed (>1.75x)** can sound slightly robotic — test on a short clip first.